Practical - 2

Practical Title: Project Initiation, Requirement Analysis, Sitemap, Wireframe, and GitHub Setup

Problem Definition: Initiate the semester-long StudentHub portal by identifying the problem scope, user roles, key modules, navigation flow, and minimum 10 pages. Create a sitemap, low-fidelity wireframe, project folder structure, README file, and GitHub repository.

A. Theory

1. Are semantic tags such as header, nav, main, section, article, aside, and footer used properly?

Answer: Header, nav, main, section, article, aside and footer are all being used the way they are meant to be used in the pages I have built so far. The header holds the page title/branding, nav wraps the menu links, and main contains the actual content of that page. I have kept only one main per page and made sure header and footer are not nested inside main, since that is what most guides recommend. Right now I have not used article or aside much because the content in a student portal like login/registration/dashboard is not really independent articles, but I plan to use them for things like the dashboard notice board. Overall the tags describe what the content actually is instead of just how it looks, which should make the site easier to read for screen readers and search engines too.
Reference: https://www.w3schools.com/html/html5_semantic_elements.asp and https://www.sitepoint.com/html/semantic-elements/

2. Are form labels, headings, image alt text, and navigation links accessible?

Answer: For the forms (login and registration) every input has a proper <label> tag connected to it, so screen readers can announce what the field is for instead of the user just seeing an empty box. Headings follow one h1 per page and go in order without skipping levels, which is something I read is important for accessibility. I have not added any images with real content yet, but wherever an image is used I am making sure to add descriptive alt text, and decorative images will get an empty alt. The navigation links in the nav bar use actual text (Home, Login, Registration etc.) instead of vague labels like "click here", so their purpose is clear from the link text alone.
Reference: https://www.w3.org/WAI/tutorials/forms/labels/ and https://webaim.org/articles/label-name/

3. Is the page structure consistent across all pages?

Answer: Yes, I have tried to keep the page structure consistent across all the pages of the site. Every page uses the same header, the same nav bar with the same links in the same

order, and the same overall layout pattern (heading, form/content, footer), just with the middle content changing per page. I linked the same stylesheet on every page too so the look and feel stays uniform. Keeping this consistent makes the site more predictable for users and also easier for me to maintain, since I am not rewriting the layout from scratch for every new page.

Reference: https://www.w3.org/TR/WCAG22/ and
https://www.computersciencecafe.com/html-stucture-and-layout.html

B. Implementation/ Output/Result

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <title>StudentHub - Login</title>
  <link rel="stylesheet" href="../css/script.css">
</head>

<body>
  <h1>Login</h1>
  <nav>
    <a href="../index.html">Home</a>
    <a href="login.html">Login</a>
    <a href="registration.html">Registration</a>
    <a href="profile.html">Profile</a>
    <a href="dashboard.html">Dashboard</a>
    <a href="course.html">Course</a>
    <a href="assignment.html">Assignment</a>
    <a href="attendance.html">Attendance</a>
    <a href="result.html">Result</a>
    <a href="contact.html">Contact</a>
  </nav>

  <h2>Student Login</h2>
  <form>
    <p>
      <label>Username:</label>
      <input type="text" name="username">
    </p>
    <p>
      <label>Password:</label>
      <input type="password" name="password">
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    </p>
    <p>
      <button type="submit">Login</button>
    </p>
  </form>

  <p>New student? <a href="registration.html">Create an account</a>.</p>
</body>

</html>
```

C. Evidence of Work (Screenshots)

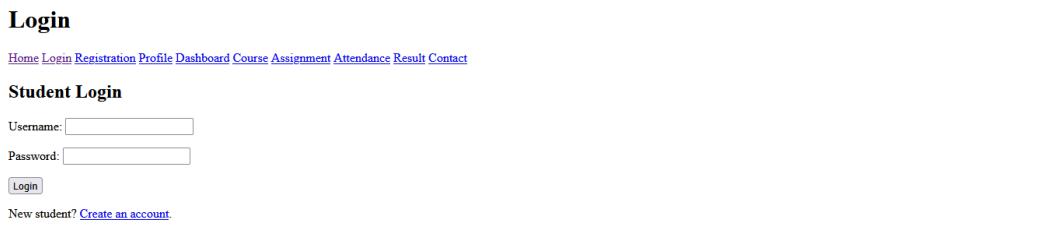


Figure 1.1: Student Login Page

Description:

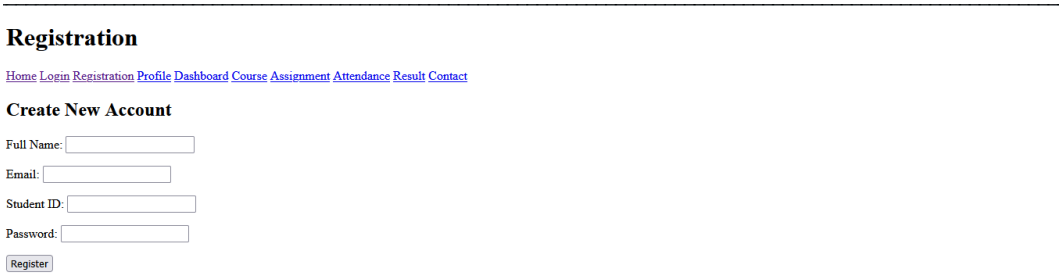


Figure 1.2: Student Registration Page

Description:

D. GitHub Repository Link:

https://github.com/VRAJ-710/studenthub-portal

E. Challenges Faced:

I had a difficulty in remembering html tags and implementing it like i knew what i wanna make but how was a issue so i learned the uses of html tags and tried to see which fits my desired page

F. Learning Outcomes:

Through this practical I learned how to plan a multi-page website before writing any code, starting with the problem definition, sitemap and a low-fidelity wireframe. I got comfortable with core HTML structure tags like header, nav, main and form elements, and understood why labelling form fields properly and keeping navigation consistent across pages matters for accessibility. I also learned the basics of setting up a project folder structure with a README and pushing the initial project to a GitHub repository, which gave me a first look at using Git for version control in a real project.

Faculty Remarks: